BUS 110: GATEWAY TO BUSINESS ANALYTICS

CASE STUDY 3

Beautiful Homes Inc.

A Microsoft Excel case for a business computing applications course

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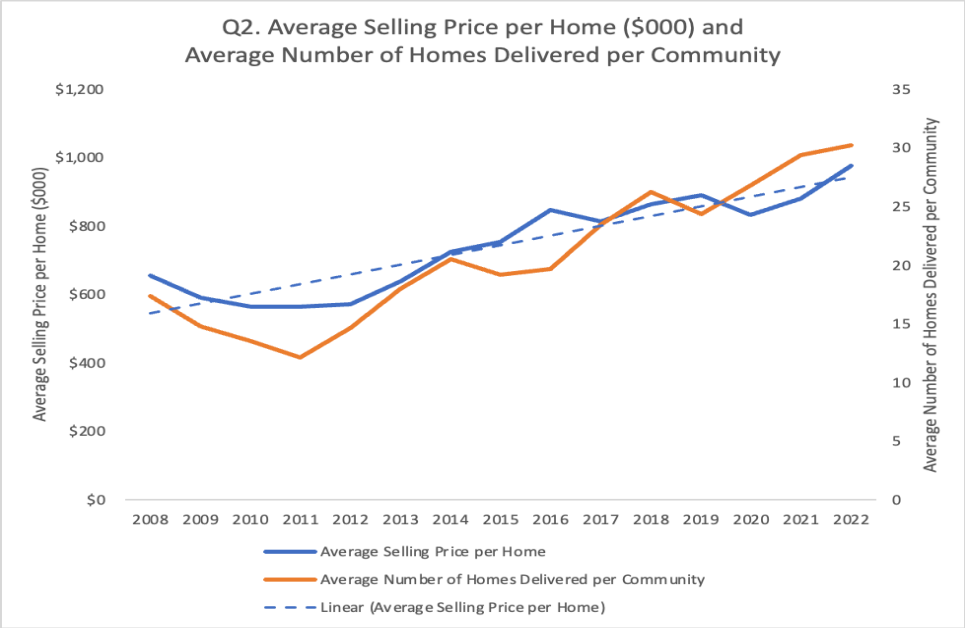
a. Beautiful Homes Inc. had the lowest average selling price per home ($000), which was $565, in 2011.

Beautiful Homes Inc. had the highest average selling price per home ($000), which was $977, in 2022.

The average selling price per home ($000) changed between the lowest year and the highest year by 73% (formula: (977-565)/565*100), or increasing $412 (formula: 977-565).

	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Average Selling Price per Home	$655	$592	$566	$565	$573	$639	$725	$755	$848	$813	$864	$891	$833	$880	$977
Average Number of Homes Delivered per Community	17	15	14	12	15	18	21	19	20	23	26	24	27	29	30

b. Line chart from Requirement #2:

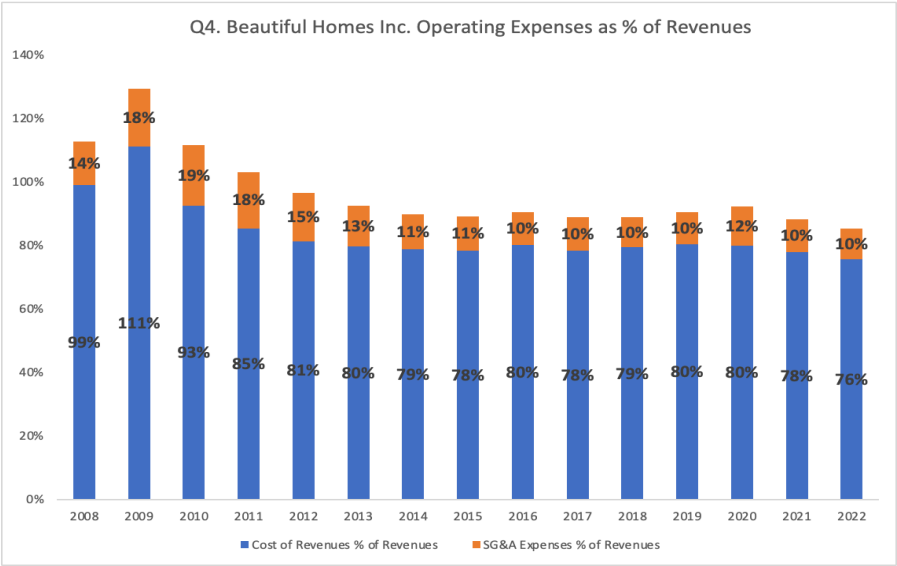


The linear trend line added to the Average Selling Price per Home ($000) data series indicates a positive relationship between time and home prices. The increasing trend suggests that, on average, home prices are rising over the period of 2008-2022. This could be influenced by factors like economic growth, increased demand, or improvements in home quality. The trend line provides a statistical representation of the overall upward trend in home prices. Moreover, we can also expect an increase in the total revenues due to the increase of Average Selling Price per Home, which means Beautiful Homes Inc. is doing a good job selling houses.

c. There is a value of Cost of Revenues % of Revenues greater than 100%, which was 111% in the year 2009. A Cost of Revenues % of Revenues value greater than 100% suggests that the company is spending more money to generate revenue than it is earning. In the context of costs versus revenues, this indicates a negative profitability scenario, possibly due to high production costs, operational inefficiencies, or unexpected expenses. The value of 111% in 2009 indicates a particularly challenging financial situation for that year, and further investigation into the company's financial statements and context is necessary to understand the underlying causes.

	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Cost of Revenues % of Revenues	99%	111%	93%	85%	81%	80%	79%	78%	80%	78%	79%	80%	80%	78%	76%
SG&A Expenses % of Revenues	14%	18%	19%	18%	15%	13%	11%	11%	10%	10%	10%	10%	12%	10%	10%

d. Stacked Column chart from Requirement #4



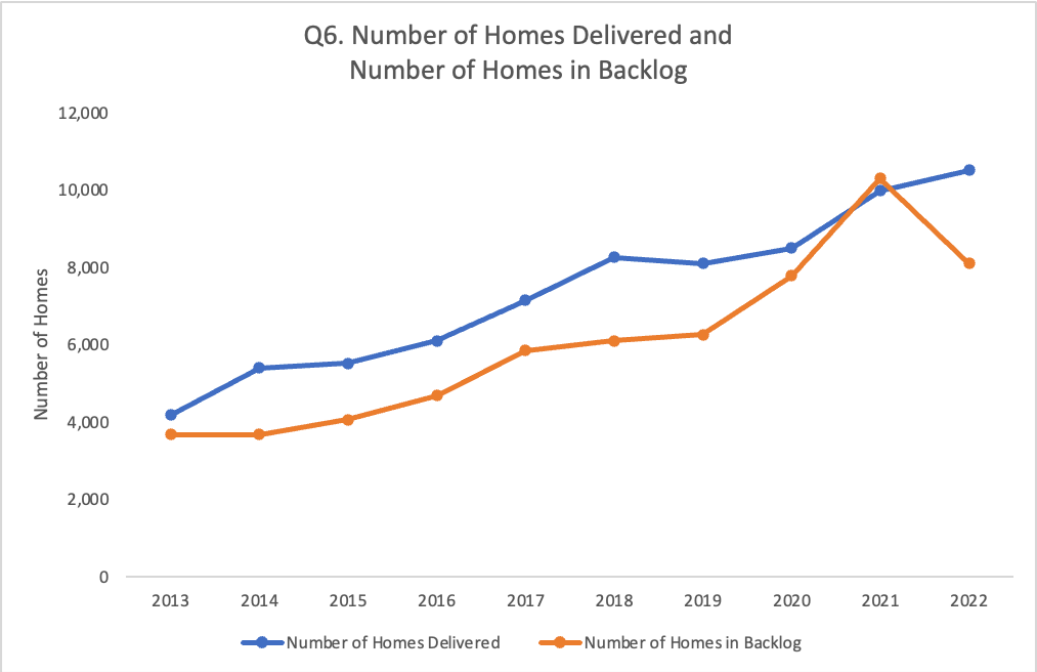
In my opinion, I would suggest adding or modifying some design elements to improve the chart's presentation of total operating expenses as a percentage of revenues as below:

- About the data labels, it looks difficult to read and likely causes overlap between numbers. Therefore, I suggest changing the color of the data labels so it looks more clearly and easily readable. If we want to keep the color as black, then we should change the colors for stacked columns, making it light orange and light blue or any lighter colors. Moreover, maybe we can also change the positions of the data labels because putting them in the centers is not best for presentation since the audience might be overwhelmed because of seeing a lot of data. Additionally, include a total label at the top of each column, indicating the total percentage of operating expenses relative to revenues.
- Furthermore, I think a Stacked Column chart may not be the best option to present this data. A line chart could be a more effective choice for depicting Beautiful Homes Inc.'s operating expenses as a percentage of revenues from 2008 to 2022. Line charts excel at illustrating trends over time, providing a clear and smooth visualization of how cost structures evolve annually. By utilizing separate lines for Cost of Revenues % of Revenues and SG&A Expenses % of Revenues, the chart can distinctly showcase the trajectory of each expense component, allowing stakeholders to discern individual trends more easily. Line charts also offer superior clarity in labeling and readability, making them particularly suitable for presenting percentages and comparisons across numerous data points. This visual approach ensures that the audience can readily interpret the data, facilitating a nuanced analysis of Beautiful Homes Inc.'s financial performance over the specified timeframe. However, Stacked Column Charts still work well when we want to show the composition of a whole (total revenues) and how its parts (expenses) contribute to it, but we need to ensure that the chart is not too cluttered, especially if we have a significant number of years.
- Lastly, I think we should also change the chart title to improve the clarity. Making it to "Total Operating Expenses as a Percentage of Revenues" might be a good option.

e. In 2021, the Number of Homes in Backlog (10,302) was growing faster than the

Number of Homes Delivered (9,986). Homes in Backlog are new homes that are under a sales contract, but are not completed/delivered to the buyer, and Homes Delivered mean new homes that have been finished and delivered to the buyer (i.e., an ownership transfer). In the period of 2013-2022, it's noteworthy that 2021 stood out as the only year when the Number of Homes in Backlog grew faster than the Number of Homes Delivered, highlighting a distinctive trend within the specified timeframe. This unique occurrence in 2021, where the backlog outpaced deliveries, suggests strong demand for the company's custom-built homes. The growing backlog indicates a robust sales pipeline and potentially higher future revenues. However, effective management of the construction process is crucial to meet this demand and ensure customer satisfaction. Overall, the data signals positive short-term prospects for Beautiful Homes Inc., with a notable emphasis on the demand for its premium single-family homes in 2021.

Row Labels	Number of Homes Delivered	Number of Homes in Backlog
2013	4,184	3,679
2014	5,397	3,679
2015	5,525	4,064
2016	6,098	4,685
2017	7,151	5,851
2018	8,265	6,105
2019	8,107	6,266
2020	8,496	7,791
2021	9,986	10,302
2022	10,515	8,098



f. Achieving the needed increase in the Number of Homes Delivered from 10,515 to 11,256 in 2022 for Beautiful Homes Inc. to reach $11 billion may be challenging. They will need to increase 741 houses. This substantial jump surpasses the previous year's increase and hinges on factors like market demand, construction capacity, and economic conditions. While historical trends indicate growth, the specific magnitude of this required increase raises feasibility concerns. External factors, such as supply chain disruptions or economic uncertainties, could pose challenges to meeting the ambitious revenue target. Careful consideration of these dynamics is essential in assessing the likelihood of reaching the desired financial milestone in 2022.

11,256	Number of Homes (2022)
$977	Average Sell Price
$11,000,000	Revenue

g. The company Toll Brothers is at risk from economic fluctuations, especially inflation, higher interest rates, and deflation. Inflation is a big threat as it raises costs for important things like land, materials, labor, and interest rates. The increase in expenses may affect housing affordability and limit the company's ability to adjust prices with

inflation, reducing profits. Inflation can lead to higher costs for the company, affecting its operations and finances. On the other hand, deflation is also a dangerous situation. Deflationary trends may lead to reduced spending and borrowing, which could harm the economy and employment. This could also devalue the company's inventories and homes. These outcomes show how the company is vulnerable to economic shifts. It emphasizes the importance of planning and managing risks to deal with these uncertainties. (page 12)

Additionally, the company's dependence on accessible financing for homebuyers increases risk. Higher mortgage rates in 2022 have affected home demand, showing a link between financing and sales. Rising mortgage rates, stricter credit standards, and more government regulation could make it harder for buyers to get home loans. This limitation may discourage homebuyers from buying or selling homes through the company. Obstacles in getting financing can be risky, hurting sales and the company's performance. Stable and accessible financing options are crucial for the company's success. (page 18)

h. Toll Brothers has demonstrated significant growth in its total assets from October 31st, 2021, to October 31st, 2022, marking an increase from $11.54 billion to $12.29 billion. This expansion suggests successful capital allocation, potentially through investments in acquisitions, infrastructure, or other strategic ventures. Additionally, the augmentation in total revenue across the same period, ascending from $8.79 billion in 2021 to $10.28 billion in 2022, reflects a robust increase in the company's top-line performance, indicating improved sales, market demand, or enhanced operational efficiency.

Moreover, the surge in net income over the past three years indicates a consistent upward trajectory, with 2022 net income reaching $1.29 billion compared to $833.63 million in 2021 and $446.62 million in 2020. This sustained growth in profitability signifies the company's ability to effectively manage costs, optimize revenue streams, or introduce innovative products/services that resonate with its customer base. The

escalation in total retained earnings from $4.97 billion in 2021 to $6.17 billion in 2022 underscores the company's dedication to preserving profits and reinvesting them for future growth or shareholder returns, highlighting a favorable financial outlook.